Ananya Vivek Kulkarni

+91 7802032289 | ananyakulkarni2103@gmail.com | ananya-kulkarni-609213244 | ananya12k | Google Scholar

Overview

MS by Research student at **CVIT, IIIT Hyderabad**, advised by Prof. C. V. Jawahar. Alongside my MS, I lead **VISIA** at INAI, IIIT Hyderabad, taking adverse-weather restoration from research to edge deployment. My research explores how adverse visual conditions such as fog, rain, and low illumination affect visual representations and downstream perception, and how image and video restoration can improve their reliability. I also work on physically grounded degradation modelling, domain adaptation, and efficient inference.

Work Experience

- **AI/ML Engineer Intern, INAI, IIIT Hyderabad** **Jun 2025 – Present**
  Supervisor: Mr. Govind Krishnan and Dr. Shankar Gangisetty
  - **Leading VISIA** (Visibility Improvement System using AI) end-to-end, owning both research and deployment of a restoration system for driving in fog, rain, and low light.
  - Built **physics-based data synthesis** to overcome the scarcity of paired real-world data: a fog pipeline grounded in atmospheric scattering (**100K+ temporally consistent frames**) and **rain modeling with ego-motion**.
  - Developed a **transmission-guided, semantic-aware video defogging algorithm** with temporal alignment for consistent restoration across varying fog densities.
  - Engineered the **end-to-end inference pipeline** with **GPU optimizations**, improving server-side efficiency while maintaining restoration accuracy.
  - Ported the **VISIA** and **TRAIL** pipelines (**22 models**) to **NVIDIA Jetson Orin Nano** for on-device edge inference.
- **Research Intern, Space Applications Centre (SAC), ISRO** **Dec 2024 – Apr 2025**
  Supervisor: Dr. Bipasha Paul Shukla
  - Designed an **ML surrogate for physics-based radiative transfer models** of greenhouse gases (CO_2, CH_4), addressing their high computational cost.
  - Generated a **9 TB** dataset on **HPC clusters** for model training and evaluation.
  - Developed **forward** (spectrum prediction) and **backward** (retrieval) models, achieving $\mathbf{R^2 = 0.994}$ and a **$\sim$750$\times$ speedup** (15+ min $\rightarrow$ 1.2 s per scenario).
- **Research Intern, Defence Institute of Advanced Technology (DIAT), DRDO** **May 2024 – Jul 2024**
  Supervisor: Dr. Amrita Nighojkar
  - Developed and evaluated **LSTM and RNN models** for leak detection in simulated maritime pipeline data, achieving over **92% accuracy**.

Publications

Enhancing Driving Visibility via Semantic-Guided Knowledge Distillation Framework for Adverse Weather Removal
H. S. Mukkamala, S. Gangisetty, **A. Kulkarni**, V. G. Yalla, and C. V. Jawahar
In Proceedings of the 16th Indian Conference on Computer Vision, Graphics and Image Processing (**ICVGIP 2025**), ACM, 2025, pp. 1–9. (DOI)

Improving Vehicle Visibility in Fog Environment
A. Kulkarni, Z. Barad, and H. B. Prajapati
In Proceedings of the 4th OPJU International Technology Conference on Smart Computing for Innovation and Advancement in Industry 5.0 (**OTCON 2025**), IEEE, 2025, pp. 1–8. (DOI)

Raspberry Pi-Driven Affordable Image-to-Braille Converter for Visually Impaired Users
A. Kulkarni, M. Shah, N. Thakur, S. Pednekar, and V. H. Shah
In *Advancements in Smart Computing and Information Security*, Springer Nature Switzerland, 2024, pp. 225–242. (DOI)
Best Paper Award, ASCIS 2023 | **Government-funded** under SSIP, Govt. of Gujarat

Gender Agreement in Indo-Aryan Languages Using Rule-Based Parsing and Finite Automata
A. Kulkarni, S. Pednekar, and D. Vegda
In *Innovations and Applications of Next-Gen Communications*, Proceedings of ERCICAM 2025, Lecture Notes in Electrical Engineering, Springer Nature Singapore, 2026, pp. 379–392. (DOI)

Manuscripts

DriveDefog: Transmission-Guided Semantic-Aware Video Defogging for Driving
A. Kulkarni, S. Gangisetty, and C. V. Jawahar

Research Projects

- **FogGod: Robust Visual SLAM under Fog and Adverse Visual Conditions** **Aug 2026 – Present**
  - Benchmarked **ORB-SLAM3, DROID-SLAM, and Rover-SLAM** on eight SubT-MRS sequences under physically modeled synthetic fog (**atmospheric scattering** with monocular depth).
  - Investigated whether image restoration improves downstream localization, linking image quality (**PSNR, SSIM**) to trajectory accuracy (**ATE RMSE**).

- **Pronunciation Error Detection with Restricted Substitutions** **Aug 2026 – Present**
  - Reproduced **PP-AF and PA-AF** CTC-based mispronunciation detection and designed data-informed phoneme substitution dictionaries, evaluated on **SpeechOcean762, L2-ARCTIC, and EpaDB**.
  - Accelerated PP-AF candidate scoring by up to **11.94**× with **92% fewer candidates** while retaining competitive detection performance.

- **RADBench-AD: Reliability Among Distractors in Autonomous Driving** **Jan 2026 – May 2026**
  - Developed a benchmark for **counterfactual referring expression comprehension** in multimodal LLMs for driving, spanning absent objects, attribute changes, and look-alike distractors.
  - Exposed failure modes of state-of-the-art multimodal models in **grounding, distractor rejection, and counterfactual reasoning**.

- **Open-Source Contribution – Graph Weather (OpenClimateFix)** **Jul 2025 – Present**
  - Implemented an **Angular MSE loss using Spherical Harmonics** to mitigate the spectral double penalty in weather forecasting, with unit tests and evaluation-pipeline integration.

Patent

- **A System and Method for Real-Time Text-to-Braille Converter** **Dec 2024**
  Indian Patent Application No. 202421066714 (published). Low-cost Braille converter with adjustable output speed, voice activation, AI image captioning, and a teacher monitoring tool.

Technical Skills

Languages	Python, C++, C, Java, SQL
Deep Learning	PyTorch, TensorFlow, Scikit-learn, Knowledge Distillation, Domain Adaptation
Computer Vision	OpenCV, Image/Video Restoration, Object Detection, Synthetic Data Generation, COLMAP, Monocular Depth Estimation, Visual SLAM
Systems & Deployment	NVIDIA Jetson, GPU Optimization, Model Quantization, Docker, SLURM, Linux, Git
Web & App Development	Django, React.js, Spring Boot, Flutter, REST APIs

Education

IIIT Hyderabad Hyderabad, India
MS by Research, Computer Science and Engineering (CVIT) **2025 – Present**

Dharmsinh Desai University Nadiad, India
B.Tech, Information Technology **2021 – 2025**